

DEF CON 34 • CAR HACKING VILLAGE BADGE • CAN HARNESS

OBD2 → DB9 cable for the CHV badge

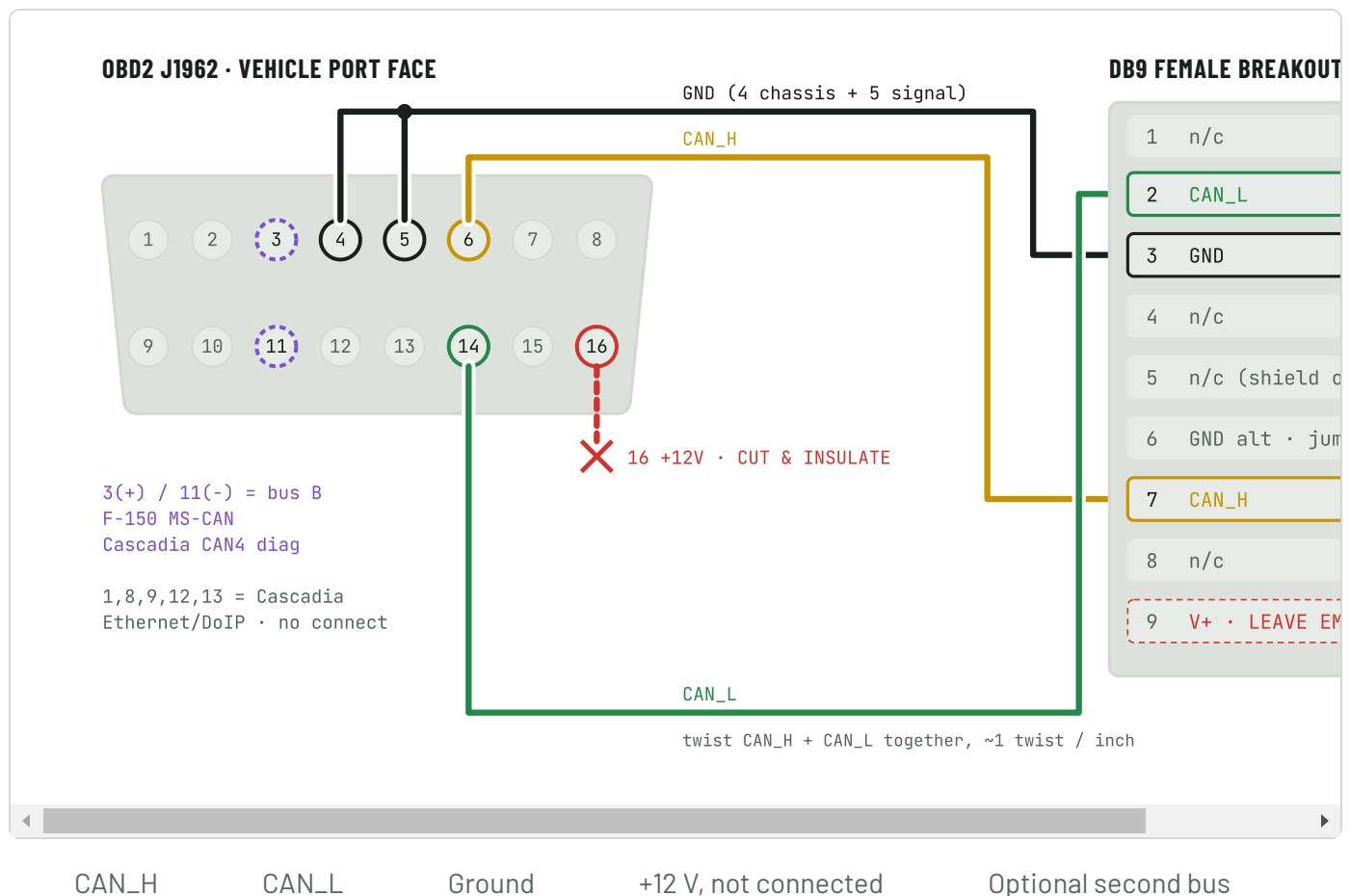
One passive cable: J1962 OBD2 pigtail on the vehicle side, DB9 female screw-terminal breakout on the badge side. CAN only. No 12 V goes to the badge.

One thing to confirm before you solder

The badge DB9 pinout isn't published in the CHV repos. This sheet assumes the common CAN-on-DB9 layout (CiA 303-1: pin 7 CAN_H, pin 2 CAN_L, pin 3 GND). Check the badge silkscreen, or ring pins 2 and 7 to the transceiver's CANH/CANL pins with a meter.

Cascadia pinouts are now taken from Freightliner drawing G66-38846 rev A (WRG-DL, J1939): -000 is the 9-pin dash connector, -001 is the 16-pin floor connector.

WIRING DIAGRAM



OBD2 face is drawn as the vehicle's socket, looking into it (pin 1 top-left). Your pigtail's plug is the mirror image and wire colors vary by seller, so ring every wire to its pin number with a meter before trusting colors. Use the numbers printed on the breakout's screw

terminals, not the DB9 face.

PIN MAP

OBD2 PIN	SIGNAL	DB9 PIN	NOTES
6	CAN_H	7	High-speed CAN. Twisted with pin 14.
14	CAN_L	2	High-speed CAN.
4 + 5	GND	3	Join chassis and signal ground. Optionally jumper DB9 3→6.
16	+12 V	—	Do not connect. The badge runs on USB. Pin 9 on many DB9 CAN tools is power in; keep it empty.
3 / 11	Bus B	7 / 2	Pin 3 = H, pin 11 = L on <i>both</i> trucks (F-150 MS-CAN, Cascadia CAN4_DIAG). Add a DPDT switch: position A = 6/14, position B = 3/11. Common terminals go to DB9 7 and 2.
1, 8, 9, 12, 13	ETH	—	Cascadia 16-pin: automotive Ethernet diag pairs + activation line. Leave unconnected.

PER-VEHICLE SETTINGS

VEHICLE / PORT	CAN_H	CAN_L	GND	BITRATE	STATUS
2017 F-150 · OBD2 HS-CAN	6	14	4, 5	500k	Standard
2017 F-150 · OBD2 MS-CAN	3	11	4, 5	125k	Ford standard
Cascadia Gen5 · 16-pin (floor, -001) · J1939	6	14	4, 5	500k*	Factory sheet
Cascadia Gen5 · 16-pin (floor, -001) · CAN4 diag	3	11	4, 5	500k?	Factory sheet · bitrate unconfirmed
Cascadia Gen5 · 9-pin (dash, -000) · J1939	C	D	A	500k*	Factory sheet
Cascadia Gen5 · 9-pin (dash, -000) · CAN4 diag	H	J	A	500k?	Factory sheet · bitrate unconfirmed

* The J1939 splices are named DL_500_..._H1939±, which points to 500k J1939. If you see only error frames, drop to 250k. CAN4 bitrate isn't on the drawing: try 500k listen-only first.

CASCADIA FACTORY PINOUT (G66-38846 REV A)

16-PIN	CIRCUIT	FUNCTION
6	H1939+ Y	J1939 CAN_H
14	H1939- GRN	J1939 CAN_L
3	508E BR-LTBL	CAN4_DIAG_H
11	508G BR-W	CAN4_DIAG_L
4, 5	GND 1204 BK	Ground
16	331 R	+12 V (F37)
1 / 9	508EE1± PRP	ETH1 diag RX P/N
12 / 13	508EE3± DKBL	ETH1 diag TX P/N
8	508EE2 DKG	Ethernet activation line

9-PIN	CIRCUIT	FUNCTION
C	H1939+ Y	J1939 CAN_H
D	H1939- GRN	J1939 CAN_L
H	508E BR-LTBL	CAN4_DIAG_H
J	508G BR-W	CAN4_DIAG_L
A	GND 1204 BK	Ground
B	331 R	+12 V (F37)
E, F, G	—	Not populated

CAN4_DIAG comes from the Central Gateway (CGW), so it's the gateway's diagnostic port, not a raw vehicle bus. Expect mostly request/response traffic rather than a busy broadcast stream. J1939 on 6/14 (C/D) is the powertrain backbone.

The OBD2 pigtail won't mate with the 9-pin Deutsch port. If your truck has the dash 9-pin, build a second short cable from a Deutsch HD10-9-1939P pigtail to an identical DB9: C→7, D→2, A→3, and optionally H/J on a DPDT switch like the 16-pin cable. B (+12 V) stays unconnected. The badge side is then the same for every vehicle.

BUILD STEPS

- 1. Confirm the breakout is passive: screw terminals only. no MAX232 or other chip on

METER CHECKS
BEFORE FIRST PLUG-IN

terminator only, no 110V or other jump on the board. "RS232" on the listing doesn't matter if it's just a straight-through breakout.

2. Ring out the pigtail. Find the wires on pins 4, 5, 6, 14 and 16 and label them with tape.
3. Cut pin 16's wire short, heat-shrink the end, and fold it back.
4. Twist 6 and 14 together. Keep the whole cable short, under about 30 cm, since it's a stub on the vehicle bus.
5. Land 6→7, 14→2, and 4+5 together→3 on the breakout terminals.
6. Leave DB9 pin 9 and the shell unconnected.

1. Cable alone: DB9 7 ↔ OBD 6, DB9 2 ↔ OBD 14, DB9 3 ↔ OBD 4 and 5 all read continuity.
2. Cable alone: OBD 16 shows no continuity to any DB9 pin.
3. Vehicle, key off: across OBD 6 and 14 you should read about 60 Ω (two 120 Ω terminators). That confirms the bus is already terminated, so turn the badge's own terminator **off** if it has one.
4. Vehicle, key on: CAN_H and CAN_L each sit near 2.5 V to ground.